



HEALTH, SAFETY & ENVIRONMENT

Health & Safety Hazards

Scope of Construction

- Construction includes:
 - Building works
 - Renovations
 - Maintenance activities
 - Demolition work
- Covered by ILO standards:
 - Safety and Health in Construction Convention 1988 (C167)
 - Safety and Health in Construction Recommendation 1988 (R175)



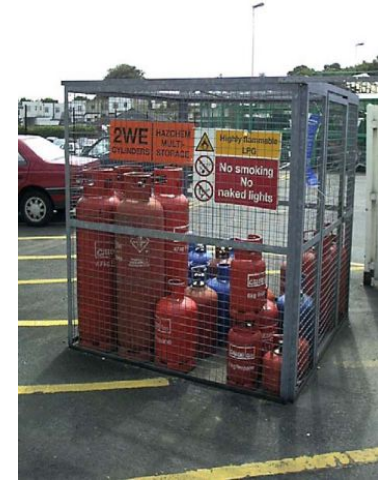
Main Construction Hazards

- Stacking of materials and flammable substances
- Storage of materials
- Storage of flammable substances
- Machinery and vehicles
- Site security
- Electricity



Storage of Materials

- Storage areas should be clearly identified
- Separate areas should be used for separate items
- Segregation of certain materials and substances
- Areas should be kept clean and tidy and routinely inspected
- Appropriate warning signs should be displayed
- Storage areas not used for other work



Stacking

- Each stack should be of one material only
- A maximum stack height must be set
- Stacks should be vertical
- Pallets should be used to keep material off the ground
- Sufficient space must be allowed around stacks for safe movement
- Stacks must be protected from being struck by people, vehicles & equipment



Storage of Flammable Substances

- Liquids, solids and gases in separate stores
- Oxygen stored separately
- External stores away from buildings, drains or excavations
- Internal stores constructed of fire-resistant materials and ventilated
- Large stores should have 2 means of escape
- Stores suitably signed
- Hot works prohibited near storage area
- Correctly rated electrical equipment
- Fire-fighting equipment
- Stores regularly inspected



Machinery

Hazards:

- Contact with moving parts
- Ejection of materials
- Noise and vibration

Controls:

- Guarding
- Maintenance
- Proper use
- Competent users
- PPE
- Supervision



Vehicles

Hazards:

- Overturning of vehicles
- Collision with pedestrians
- Collision with structures



Controls:

- Site layout
- Routes for pedestrians/vehicles
- Warning signs
- Maintenance, proper use
- Competent drivers
- Movements supervised e.g. banksman for reversing

Forklift Trucks

- Collision with structures and pedestrians
- Obscured visibility especially when reversing
- Noise and vibration
- Overturning



Dumper Trucks

- Collision with structures and pedestrians
- Overturning on slopes and edge of excavations
- Poorly maintained braking systems
- Unauthorised use



Cement Mixers

- Entanglement with rotating drum, internal blades
- Entrapment by belt drives
- Dry cement dust and wet cement burns
- Exposure to fuel vapours
- Noise and vibration
- Overturning



Site Security

ISSUES

- Public taking short cuts
- Public right of way
- Thieves
- Children
- Perimeter fence and signs
- Security gates
- Security staff
- Lighting

MITIGATION

- CCTV cameras and/or site alarm
- Secure all portable equipment
- Remove ladders from scaffolds
- Secure all chemicals
- Secure all mobile plant
- Cover or barrier off excavations

What Control Measures Can You See?



Electricity

Hazards specific to construction:

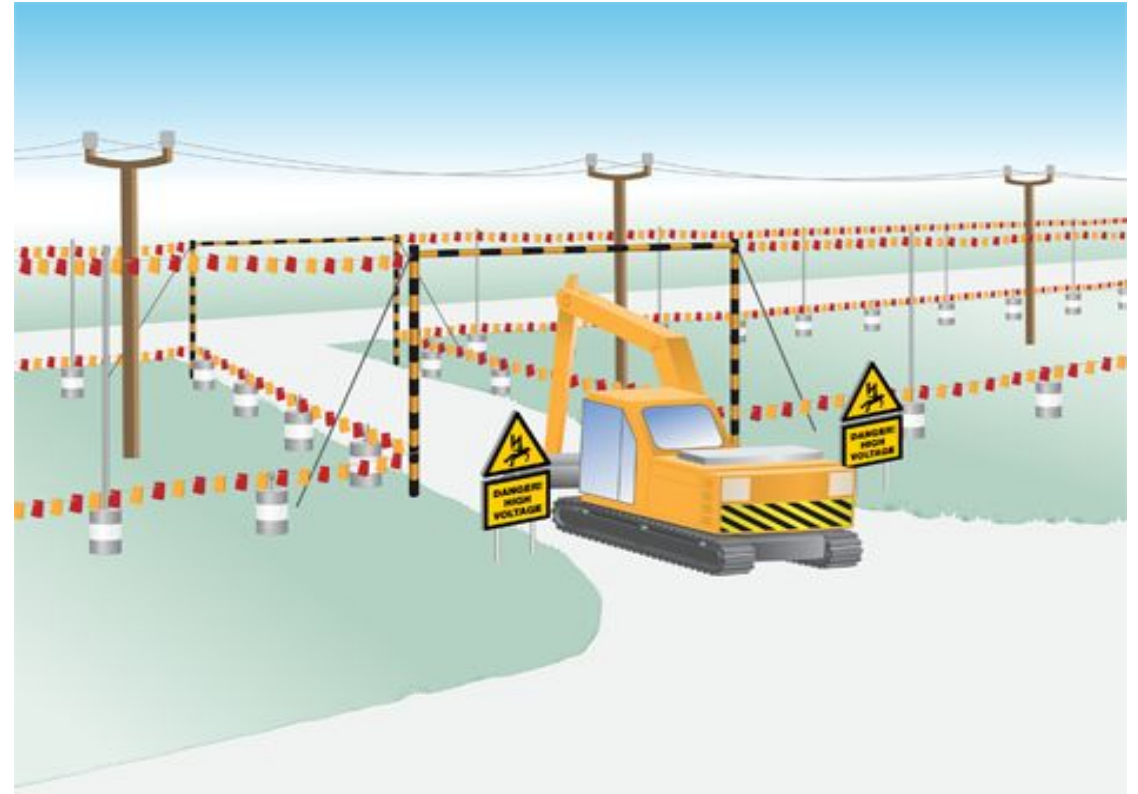
- Contact with overhead power lines
- Contact with buried services

Don't forget, direct contact isn't necessary: electricity can **arc**



Overhead Power Line – Precautions

- Isolating the power supply
- Using Safe-Systems-of-Work and Work-Permits
- Sleeve low voltage power lines
- Using barriers, signage and goal-posts
- Using banksmen
- Using non-conducting equipment



Demolition

Includes:

- Total destruction
- Dismantling of structures



Hazards:

- Premature collapse
- Work at height
- Plant, vehicles, machinery
- Live overheads
- Buried services
- Asbestos
- Dust
- Explosives
- Biological hazards
- Sharps
- Manual handling

Demolition Controls

- Choice of method to exclude people
- Structural surveys
- Disconnection of services
- Removal and disposal of hazardous materials
- Security for the site
- Camping down to minimise dust
- **NB controls for heights and excavations considered separately later**

Working at Height

Work at any height where there is a risk of a fall liable to cause personal injury:

- Steel workers, scaffolders, roofers, engineers, welders, maintenance staff, painters, window cleaners
- Main risks:
 - worker falling
 - object falling
- Accidents:
 - death
 - neck or spinal injury
 - broken bones



Risk Factors

- Deterioration of materials
- Unprotected edges
- Unstable access equipment
- Weather
- Falling materials
 - crumbling brickwork, loose tiles
 - bad storage of materials on scaffolding
 - gaps in platform surfaces
 - open, unprotected edges
 - incorrect methods of getting materials to and from roof



Control Measures

Control measures will depend upon:

- Nature and duration of task
- Competence
- Training
- Planning and supervision
- Suitability of equipment, maintenance
- Working platforms
- PPE, e.g. harnesses and helmets
- Fall arrest systems
- Weather
- Health of workers
- Compliance with regulations



Ladder Controls

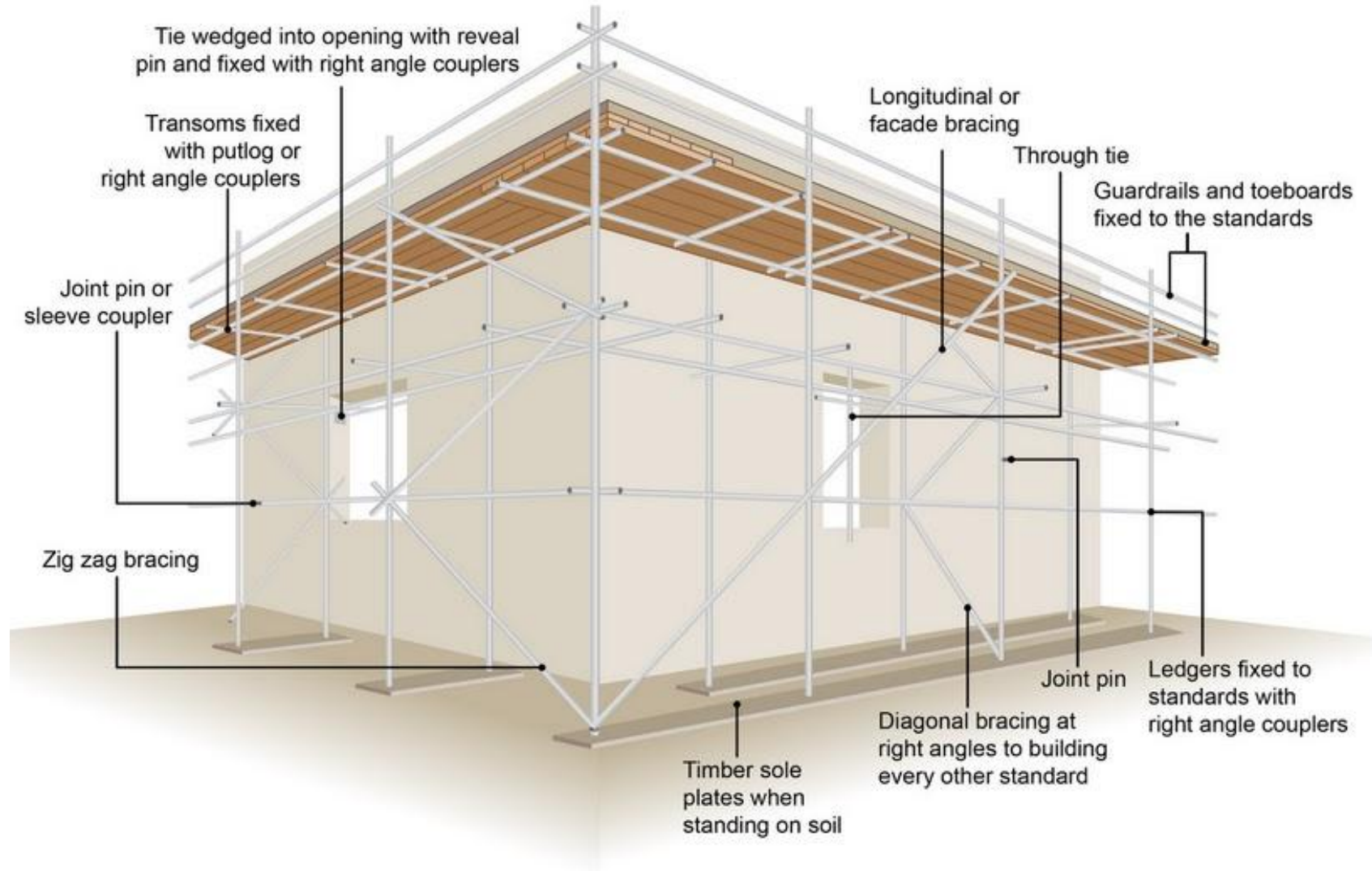
- Sited away from live overheads
- Solid, flat base
- Hands on stiles, never on rungs
- Correct angle (1:4 rule – 75°)
- Top of the ladder against solid support
- Ladder secured at the top, or:
 - guy ropes attached or,
 - ladder should be “footed”



Basic Components of Independent Tied Scaffolds

Standards	– Uprights or vertical tubes
Ledgers	– Horizontal tubes
Transoms	– Short horizontal tubes spanning across ledgers, perpendicular to building face, used to support work platform
Bracing	– Diagonal tubes for supporting standards & ledgers
Base plates	– Metal plates put underneath the standards to prevent them from sinking in the ground
Sole boards	– Timber under base plates
Work platform	– Fully boarded
Guardrails	– Enclose work platform
Toe-boards	– Provide lip to platform

Independent Tied Scaffold



Mechanical and Non-Mechanical Hazards

Mechanical hazards

- Contact with or being caught up in moving parts

Non-Mechanical hazards

- From power source or things being emitted by the machine



Mechanical Hazards

- Crushing
- Shearing
- Cutting or severing
- Entanglement
- Drawing in or trapping
- Impact
- Stabbing or puncture
- Friction or abrasion
- High pressure fluid injection



Non-Mechanical Hazards

- Electricity
- Noise
- Vibration
- Hazardous substances
- Ionising radiation
- Non-ionising radiation
- Extreme temperatures
- Ergonomics
- Slips, trips and falls
- Fire and explosion



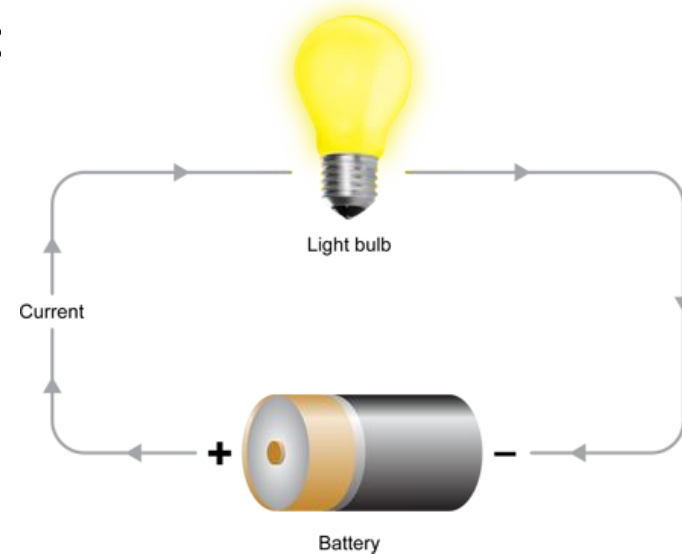
Machinery Guards

- Fixed guards
- Interlocked guards
- Adjustable/self-adjusting guards
- Trip devices
- **Other Protective Measures**
 - Two-handed controls
 - Protective appliances
 - Emergency stop controls
 - Personal protective equipment
 - Information, instruction, training and supervision



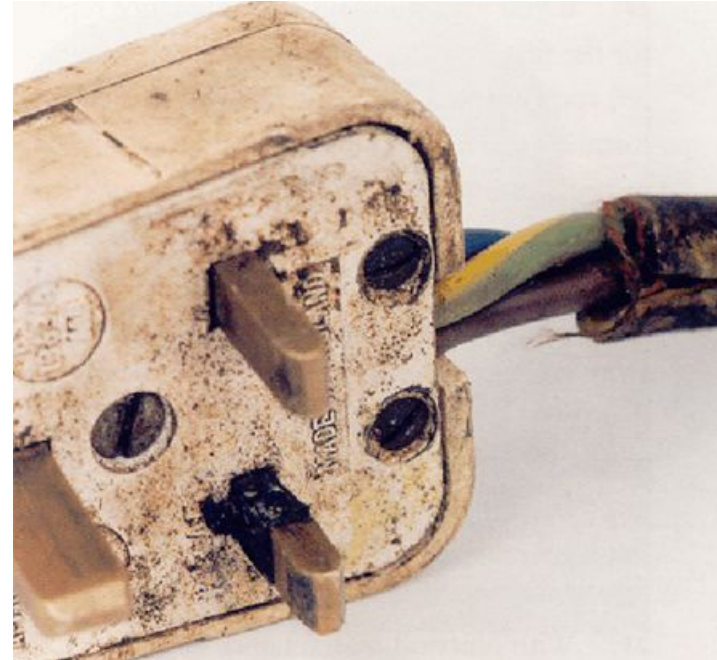
Basic Principles

- An **electrical circuit** comprises 3 elements:
- **Pressure** = potential difference
 - Measured in **volts** (V)
- **Flow rate** = current
 - Measured in **amps** (I)
- **Resistance** = obstruction in circuit
 - Measured in **ohms** (R)
- Linked together by **Ohm's Law**:
$$V = I \times R$$



The Hazards of Electricity

- Electric shock
 - Heart fibrillation
 - Muscle Contraction
 - Burns
 - Arcing
- Fire and explosion
- Secondary effects



Severity of Shock Influenced By:

- **Voltage** – the higher the voltage, the greater the current
- **Duration** – the length of time that a person is exposed
- **Current path** – the route that the electricity takes through the body
- **Frequency** – of the AC current
- **Resistance** – skin condition, clothing, etc.
- **Contact surface area** – the more skin that is in contact, the more severe the injury
- **Environmental factors** – metal surfaces, humidity, etc.

Protective Systems

- Fuses
- Earthing
- Isolation of supply
- Reduced and low voltage systems
- Residual Current Devices (RCD's)
- Double insulation

What is Fire?



Fire is a rapid chemical process in which oxygen combines with another substance in the presence of a source of heat

This reaction is called **combustion**

During this reaction **heat, flames** and **smoke** are produced

The Fire Triangle



Classification of Fires

Class A	- solid materials
Class B	- liquids and liquefiable solids
Class C	- gases
Class D	- metals
Class F	- high temperature fat



Common Causes of Fires

- Electrical equipment
- Deliberate ignition (arson)
- Hot work
- Smoking
- Cooking appliances
- Heating appliances
- Unsafe use & storage of flammable liquids and gases
- Mechanical heat
- Chemical reactions



Fire Risk Assessment

Step 1

- Identify the fire hazards

Step 2

- Identify the people who might be harmed

Step 3

- Identify and implement the fire precautions that are required

Step 4

- Record findings, plan and train

Step 5

- Review and revise as necessary

Fire Prevention

Control flammable substances:

- Solids, liquids, gases

Control ignition sources:

- Systems of work, smoking, arson

Systems of work:

- Hot processes
- Machinery
- Electrical equipment

Good Housekeeping:

- General tidiness
- Waste control and disposal



Fire Detection

- Smoke detectors

- Detect small smoke particles
- Two main types: ionising and optical
- Can give rise to false alarms



- Heat detectors

- Detect excess heat generated by a fire
- Usually less sensitive and give later warning
- Two main types: rate of rise and fixed temperature
- May not detect fires that are giving off smoke but not much heat

Portable Fire Extinguishers

Extinguisher	Class of Fire
Water	A
Carbon dioxide	A and B
Foam	A and B
Dry powder	All



Means of Escape

- Available to every person in the workplace
- No use of lifts
- Must take staff to a place of safety
- Two or more separate routes may be required
- Travel distance should be short
- Adequate width
- Clearly signed
- Appropriately lit
- Emergency lighting
- No obstructions



Assembly Points

- Safe distance from building
- Safe location
- Further escape possible if needed
- Must not impede fire-fighters
- Clearly signed
- 'Refuges' for disabled workers



Physical Forms of Chemicals

- The physical form greatly affects the hazard presented and the route of entry into the body
 - Dust
 - Fume
 - Gas
 - Mist
 - Vapour
 - Liquids



Classification of Chemicals Hazardous to Health



Toxic



Harmful (Xn)



Carcinogenic



Respiratory Sensitiser



Irritant (Xi)



Corrosive

The Principles of Good Practice

1. Minimise emission, release and spread of hazardous substances
2. Choose effective and reliable controls
3. Use control measures proportionate to the risk
4. Use PPE where control cannot be achieved by other means
5. Implement personal hygiene and protection regimes
6. Carry out health and medical surveillance
7. Check control measures regularly
8. Additional controls for carcinogens, asthmagens or mutagens

Common Methods of Control

- Elimination or substitution
- Process change
- Reduce exposure times
- Enclosure or segregation
- Local Exhaust Ventilation
- PPE
- Personal hygiene and protection
- Health Surveillance/monitoring



Use of Signs and PPE:

